

broadcast-underlay

To configure the underlay in a LISP network to use a multicast group to send encapsulated broadcast packets and link local multicast packets, use the **broadcast-underlay** command in the service submode. To remove the broadcast functionality, use the **no** form of this command.

broadcast-underlay *multicast-ip*

no broadcast-underlay *multicast-ip*

Syntax Description	<i>multicast-ip</i> IP address of the multicast group that sends the encapsulated broadcast packets	
Command Default	None.	
Command Modes	LISP Instance Service Ethernet (router-lisp-inst-serv-eth) LISP Service Ethernet (router-lisp-serv-eth)	
Command History	Release	Modification
	Cisco IOS XE Everest 16.6.1	This command was introduced.
Usage Guidelines	Use this command to enable the broadcast functionality on the fabric edge node in a LISP network. Ensure that this command is used in the router-lisp-service-ethernet mode or router-lisp-instance-service-ethernet mode.	

Example

The following example shows how to configure broadcast on a fabric edge node:

```
device(config)#router lisp
device(config-router-lisp)#instance-id 3
device(config-router-lisp-inst)#service ethernet
device(config-router-lisp-inst-serv-eth)#eid-table vlan 250
device(config-router-lisp-inst-serv-eth)#broadcast-underlay 225.1.1.1
device(config-router-lisp-inst-serv-eth)#database-mapping mac locator-set rloc2
device(config-router-lisp-inst-serv-eth)#exit-service-ethernet
```

database-mapping

To configure an IPv4 or IPv6 endpoint identifier-to-routing locator (EID-to-RLOC) mapping relationship and an associated traffic policy for Location Identifier Separation Protocol (LISP), use the **database-mapping** command in LISP EID-table configuration mode. To remove the configured database mapping, use the **no** form of this command.

```
database-mapping eid-prefix / prefix-length { locator-set RLOC-name [ proxy | default-etr | default-etr-route-map | route-tag | silent-host-detection ] | ipv6-interface interface-name | ipv4-interface interface-name | auto-discover-rlocs | limit }
```

```
no database-mapping eid-prefix / prefix-length { locator-set RLOC-name [ proxy | default-etr | default-etr-route-map | route-tag | silent-host-detection ] | ipv6-interface interface-name | ipv4-interface interface-name | auto-discover-rlocs | limit }
```

Syntax Description

<i>eid-prefix / prefix-length</i>	IPv4 or IPv6 endpoint identifier prefix and length that is advertised by the router.
locator-set <i>RLOC-name</i>	Routing Locator (RLOC) associated with the value specified for the <i>eid-prefix</i> . Use the following keyword options for database mapping: • proxy: Enables configuration of static proxy database mapping. • default-etr: Enables configuration of default Egress Tunnel Router (ETR) database mapping. • route-tag <i>route-tag</i>: Monitors the RIB entry for a match with the <i>route-tag</i> specified. • default-etr-route-map <i>route-map</i>: Enables the route map to look for default-etr RIB route updates, and dynamically changes the locator set for this database mapping. • silent-host-detection: Enables silent host detection for hosts under the EID prefix.
ipv4 interface <i>interface-name</i>	IPv4 address and name of the interface that is used as the RLOC for the EID prefix.
ipv6 interface <i>interface-name</i>	IPv6 address and name of the interface that is used as the RLOC for the EID prefix.
auto-discover-rlocs	Configures the ETR to discover the locators of all the routers configured to function as both an ETR and an Ingress Tunnel Router (ITR)—such routers are referred to as xTRs—in the ETR LISP site when the site uses multiple xTRs, and each xTR is configured to use DHCP-learned locators or configured with only its own locators.
limit	Specifies the maximum size of local EID prefixes database.

Command Default

No LISP database entries are defined.

Command Modes LISP-instance-service (router-lisp-instance-service)

Command History	Release	Modification
	Cisco IOS XE Everest 16.6.1	This command was introduced.
	Cisco IOS XE Fuji 16.9.1	This command was modified. Introduced support for the keyword proxy .
	Cisco IOS XE Bengaluru 17.5.1	This command was modified. Introduced support for the keyword default-etr-route-map .
	Cisco IOS XE Dublin 17.11.1	This command was modified. Introduced support for the keyword silent-host-detection .

Usage Guidelines In the LISP-instance-service configuration mode, the **database-mapping** command configures LISP database parameters with a specified IPv4 or IPv6 EID prefix block. The *locator* is the IPv4 or IPv6 address of any interface used as the RLOC address for the *eid-prefix* assigned to the site but can also be the loopback address of the interface.

When a LISP site has multiple locators associated with the same EID prefix block, multiple **database-mapping** commands are used to configure all of the locators for a given EID prefix block.

In a MultiSite scenario, the LISP border node advertises about the site EID that it is attached to on the transit map server to attract site traffic. To advertise, the border node has to obtain the route from the internal border and proxy register with the transit site map server accordingly. The **database-mapping eid-prefix locator-set RLOC-name proxy** command enables the configuration of a static proxy database mapping.

In Cisco IOS XE Bengaluru 17.5.1 and later releases, the **database-mapping eid-prefix locator-set RLOC-name default-etr-route-map route-map** command monitors the specified *route-map* for route updates corresponding to the *eid-prefix*. If there is an update from the route map, and if the route map has a defined LISP locator set, the **locator-set** of this database mapping is changed to the one specified in the *route-map*.

By default, RIB metric (BGP MED attribute) information for the specified **default-etr eid-prefix** is obtained. You can disable the default, by using the **default-etr disable-metric** command.

Enabling the **default-etr-route-map** option allows you to match other BGP attributes such as AS_PATH, COMMUNITIES, and so on, and modify the locator set of the database mapping accordingly.

If the **silent-host-detection** option is enabled, LISP triggers silent host detection for the traffic destined for the host that is not present in the site table. SISF then sends probes for silent hosts, if any, in the network, and if a response is received, a site entry is added to the LISP database and traffic is forwarded accordingly.

Examples

The following example shows how to map the EID prefix with the locator-set RLOC, in the EID configuration mode on an external border. Ensure that the locator-set RLOC is already configured.

```
device(config)# router lisp
device(config-router-lisp)# instance-id 3
device(config-router-lisp-inst)# service ipv4
device(config-router-lisp-inst-serv-ipv4)#eid-table vrf red
device(config-router-lisp-inst-serv-ipv4-eid-table)# database-mapping 172.168.0.0/16
locator-set RLOC proxy
device(config-router-lisp-inst-serv-ipv4-eid-table)# database-mapping 173.168.0.0/16
locator-set RLOC proxy
device(config-router-lisp-inst-serv-ipv4-eid-table)# map-cache 0.0.0.0/0
map-requestdevice(config-router-lisp-inst-serv-ipv4-eid-table)#exit
device(config-router-lisp-inst-serv-ipv4)#
```

The following example shows how to dynamically change the EID prefix and locator set RLOC mapping, using the **default-etr-route-map** keyword:

```
device(config)# router lisp
device(config-router-lisp)# instance-id 1
device(config-router-lisp-inst)# service ipv4
device(config-router-lisp-inst-serv-ipv4)#eid-table default
device(config-router-lisp-inst-serv-ipv4-eid-table)# database-mapping 0.0.0.0/0 locator-set
RLOC default-etr-route-map abc
device(config-router-lisp-inst-serv-ipv4-eid-table)#exit
device(config-router-lisp-inst-serv-ipv4)#
```

Related Commands

Command	Description
eid-table vrf <i>vrf-name</i>	Associates the instance-service instantiation with a virtual routing and forwarding (VRF) table or default table through which the endpoint identifier address space is reachable.

first-packet-petr

To prevent the loss of the first packet (and subsequent packets until map-cache is resolved), use the **first-packet-petr** command on the Map Server, in the LISP-service or the LISP-instance-service configuration mode. To disable the configuration of this command, use its **no** form.

Configuring this command ensures that even the first packet that is sent out from the fabric edge device reaches its destination through a first-packet-handler border that is available.

```
first-packet-petr remote-locator-set fpetr-RLOC
```

```
no first-packet-petr remote-locator-set fpetr-RLOC
```

Syntax Description	remote-locator-set <i>fpetr-RLOC</i>	Specifies a remote locator-set, which is a set of IP addresses of remote devices, that connect to an external network or to networks across sites or to Data Center through remote or local sites.
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Command Default	None.
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Command Modes	LISP-instance-service LISP-service
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Command History	Release	Modification
	Cisco IOS XE Amsterdam 17.3.1	The command was introduced.

Usage Guidelines	The ITR or the fabric edge device drops the initial packets sent to it until it learns the destination EID reachability from the local MSMR. To prevent the drop of the first packet, configure the first-packet-petr command on the local MSMR. Configure the first-packet-petr command on the local map server to ensure that when the fabric edges boots up and resolves the 0/0 map-cache entry, it gets the first packet forwarding RLOCs. When an MSMR receives a request to connect to an external network (like internet), it first checks for the availability of an external border. If the map server does not find the default-ETR border or the internet service providing border, it responds with the remote RLOCs that are configured with the first-packet-petr command.
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Note	You can configure the first-packet-petr command only on a control plane that is within a fabric site. You cannot configure this command on the control plane of a transit site.
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Examples

The following example first defines a remote locator set and associates the remote RLOCs with the first-packet-petr command:

```
Device(config)#router lisp
Device(config-router-lisp)#remote-locator-set fpetr
Device(config-router-lisp-remote-locator-set)#23.23.23.23 priority 1 weight 1
Device(config-router-lisp-remote-locator-set)#24.24.24.24 priority 1 weight 1
Device(config-router-lisp-remote-locator-set)#exit-remote-locator-set

Device(config-router-lisp)#service ipv4
Device(config-lisp-srv-ipv4)#first-packet-petr remote-locator-set fpetr
Device(config-lisp-srv-ipv4)#map-server
Device(config-lisp-srv-ipv4)#map-resolver
Device(config-lisp-srv-ipv4)#exit-service-ipv4
Device(config-router-lisp)#
```

The configured behavior is inherited by all instances under service ipv4.

To override the behavior for a particular instance, configure the first-packet-petr command for that instance. In the following example, instance 101 disables the first-packet-petr command.

```
Device(config-router-lisp)#instance-id 101
Device(config-router-lisp-inst)#service ipv4
Device(config-router-lisp-inst-service-ipv4)#no first-packet-petr remote-locator-set
Device(config-router-lisp-inst-service-ipv4)#exit-service-ipv4
```

show lisp instance-id ipv4 database

To display the operational status of the IPv4 address family and the database mappings on a device, use the **show lisp instance-id ipv4 database** command in privileged EXEC mode.

show lisp instance-id *instance-id* **ipv4 database** [**silent-host-detection**]

Syntax Description	silent-host-detection (Optional) Displays the silent host detection database entry on the tunnel router (xTR).								
Command Default	None								
Command Modes	Privileged EXEC (#)								
Command History	<table> <tr> <th>Release</th><th>Modification</th></tr> <tr> <td>Cisco IOS XE Everest 16.5.1a</td><td>This command was introduced.</td></tr> <tr> <td>Cisco IOS XE Fuji 16.9.1</td><td>This command was modified. Introduced support for display of proxy database size.</td></tr> <tr> <td>Cisco IOS XE Dublin 17.11.1</td><td>This command was modified. Introduced support for the silent-host-detection keyword.</td></tr> </table>	Release	Modification	Cisco IOS XE Everest 16.5.1a	This command was introduced.	Cisco IOS XE Fuji 16.9.1	This command was modified. Introduced support for display of proxy database size.	Cisco IOS XE Dublin 17.11.1	This command was modified. Introduced support for the silent-host-detection keyword.
Release	Modification								
Cisco IOS XE Everest 16.5.1a	This command was introduced.								
Cisco IOS XE Fuji 16.9.1	This command was modified. Introduced support for display of proxy database size.								
Cisco IOS XE Dublin 17.11.1	This command was modified. Introduced support for the silent-host-detection keyword.								

Examples

The following is a sample output of the **show lisp instance-id id ipv4 database** command, displaying the EID prefixes configured for a site:

```
device# show lisp instance-id 101 ipv4 database

LISP ETR IPv4 Mapping Database for EID-table vrf red (IID 101), LSBs: 0x1
Entries total 1, no-route 0, inactive 0

172.168.0.0/16, locator-set RLOC, proxy
  Locator      Pri/Wgt  Source      State
  100.110.110.110  1/100  cfg-intf    site-self, reachable

device#

device# show lisp instance-id 101 ipv4

Instance ID:                  101
Router-lisp ID:                0
Locator table:                 default
EID table:                     vrf red
Ingress Tunnel Router (ITR):   disabled
Egress Tunnel Router (ETR):    enabled
Proxy-ITR Router (PITR):      enabled RLOCs: 100.110.110.110
Proxy-ETR Router (PETR):      disabled
NAT-traversal Router (NAT-RTR): disabled
Mobility First-Hop Router:     disabled
Map Server (MS):               enabled
Map Resolver (MR):             enabled
Mr-use-petr:                   enabled
Mr-use-petr locator set name:  site2
Delegated Database Tree (DDT): disabled
Site Registration Limit:       0
```

show lisp instance-id ipv4 database

```

Map-Request source:                derived from EID destination
ITR Map-Resolver(s):              100.77.77.77
                                   100.78.78.78
                                   100.110.110.110 prefix-list site2
ETR Map-Server(s):                100.77.77.77 (11:25:01)
                                   100.78.78.78 (11:25:01)
xTR-ID:                           0xB843200A-0x4566BFC9-0xDAA75B2D-0x8FBE69B0
site-ID:                          unspecified
ITR local RLOC (last resort):      100.110.110.110
ITR Solicit Map Request (SMR):     accept and process
  Max SMRs per map-cache entry:    8 more specifics
  Multiple SMR suppression time:    20 secs
ETR accept mapping data:           disabled, verify disabled
ETR map-cache TTL:                 1d00h
Locator Status Algorithms:
  RLOC-probe algorithm:             disabled
  RLOC-probe on route change:       N/A (periodic probing disabled)
  RLOC-probe on member change:      disabled
  LSB reports:                     process
  IPv4 RLOC minimum mask length:    /0
  IPv6 RLOC minimum mask length:    /0
Map-cache:
  Static mappings configured:       1
  Map-cache size/limit:             1/32768
  Imported route count/limit:       0/5000
  Map-cache activity check period:  60 secs
  Map-cache FIB updates:            established
  Persistent map-cache:             disabled
Database:
  Total database mapping size:      1
  static database size/limit:       1/65535
  dynamic database size/limit:      0/65535
  route-import database size/limit: 0/5000
  import-site-reg database size/limit 0/65535
  proxy database size:              1
  Inactive (deconfig/away) size:    0
Encapsulation type:                 vxlan

```

The following is a sample output of the **show lisp instance-id id ipv4 database silent-host-detection** command, displaying the silent host detection database entry on the xTR:

```
device# show lisp instance-id 101 ipv4 database silent-host-detection
```

```

LISP ETR IPv4 Mapping Database for LISP 0 EID-table vrf red (IID 101), LSBs: 0x1
Entries total 2, no-route 0, inactive 0, do-not-register 0

```

```

10.168.0.0/16, inherited from default locator-set RLOC
  Uptime: 1d10h, Last-change: 1d10h, Last-Silent-Host-Probe: 1d02h
  Domain-ID: local
  Service-Insertion: N/A
  Locator      Pri/Wgt  Source      State
  10.11.11.11  50/50      cfg-intf    site-self, reachable
10.169.0.0/16, inherited from default locator-set RLOC
  Uptime: 2d11h, Last-change: 2d11h, Last-Silent-Host-Probe: never
  Domain-ID: local
  Service-Insertion: N/A
  Locator      Pri/Wgt  Source      State
  10.11.11.11  50/50      cfg-intf    site-self, reachable

```

show platform hardware fed switch active fwd-asic resource tcam utilization

To display CAM utilization information for ASIC, use the **show platform hardware fed switch active fwd-asic resource tcam utilization** command in privileged EXEC mode.

show platform hardware fed switch active fwd-asic resource tcam utilization [*asic-number*] [*slice-id*]

Syntax Description	asic-number	Displays the ASIC number. Valid values are from 0 to 7.
	slice-id	Displays per slice usage.
Command Modes	Privileged EXEC (#)	
Command History	Release	Modification
	Cisco IOS XE Everest 16.5.1a	This command was introduced.

Examples

The following is sample output from the **show platform hardware fed switch active fwd-asic resource tcam utilization** command:

```
Device# enable
Device# show platform hardware fed switch active fwd-asic resource tcam utilization

CAM Utilization for ASIC  [0]
Table      Subtype      Dir      Max      Used      %Used      V4      V6
MPLS      Other
-----
Mac Address Table      EM      I      32768      25      0.08%      0      0
0      25
Mac Address Table      TCAM      I      1024      22      2.15%      0      0
0      22
L3 Multicast      EM      I      8192      0      0.00%      0      0
0      0
L3 Multicast      TCAM      I      512      9      1.76%      3      6
0      0
L2 Multicast      EM      I      8192      0      0.00%      0      0
0      0
L2 Multicast      TCAM      I      512      11      2.15%      3      8
0      0
IP Route Table      EM      I      24576      14      0.06%      13      0
1      0
IP Route Table      TCAM      I      8192      30      0.37%      11      16
2      1
QOS ACL      TCAM      IO      5120      85      1.66%      28      38
0      19
0      10      TCAM      I      45      0.88%      15      20
0      9      TCAM      O      40      0.78%      13      18
```

show platform hardware fed switch active fwd-asic resource tcam utilization

Security ACL	TCAM	IO	5120	131	2.56%	26	60
0 45							
0 40	TCAM	I		88	1.72%	12	36
0 5	TCAM	O		43	0.84%	14	24
Netflow ACL	TCAM	I	256	6	2.34%	2	2
0 2							
PBR ACL	TCAM	I	1024	36	3.52%	30	6
0 0							
Netflow ACL	TCAM	O	768	6	0.78%	2	2
0 2							
Flow SPAN ACL	TCAM	IO	1024	13	1.27%	3	6
0 4							
0 2	TCAM	I		5	0.49%	1	2
0 2	TCAM	O		8	0.78%	2	4
Control Plane	TCAM	I	512	290	56.64%	138	106
0 46							
Tunnel Termination	TCAM	I	512	22	4.30%	9	13
0 0							
Lisp Inst Mapping	TCAM	I	2048	2	0.10%	0	0
0 2							
Security Association	TCAM	I	256	4	1.56%	2	2
0 0							
CTS Cell Matrix/VPN	EM	O	8192	0	0.00%	0	0
Label							
0 0							
CTS Cell Matrix/VPN	TCAM	O	512	1	0.20%	0	0
Label							
0 1							
Client Table	EM	I	4096	0	0.00%	0	0
0 0							
Client Table	TCAM	I	256	0	0.00%	0	0
0 0							
Input Group LE	TCAM	I	1024	0	0.00%	0	0
0 0							
Output Group LE	TCAM	O	1024	0	0.00%	0	0
0 0							
Macsec SPD	TCAM	I	256	2	0.78%	0	0
0 2							

Output fields are self-explanatory.

Related Commands

Command	Description
show platform hardware fed switch active fwd-asic resource tcam table	Displays the current CAM table.
show platform hardware fed switch active fwd-asic resource tcam usage	Displays the current CAM usage.

Keepalive (template)

To enable keepalive timer for interface templates, use the **keepalive timer** in template configuration mode. To disable the keepalive timer, use the **no** form of this command.

keepalive *seconds*
no keepalive *seconds*

Syntax Description	<i>seconds</i> Sets the keepalive timer in seconds. The range is from 0 to 32767. Default is 10.
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Command Default	The keepalive timer is not set.
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Command Modes	Template configuration (config-template)
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Command History	Release	Modification
	15.2(2)E	This command is introduced.
	Cisco IOS XE Release 3.6E	This command is supported on Cisco IOS XE Release 3.6E.

The following example shows how to configure keepalive timer for interface templates.

```
Device# configure terminal
Device(config)# template user-template1
Device(config-template)# keepalive 100
Device(config-template)# end
```

Related Commands	Command	Description
	hold-queue	Limits the length of the IP output queue on an interface or an interface template.